

# CSA1715-ARTIFICIAL INTELLIGENCE

## LAB PROGRAM- 15

06-08-2026

15. Write the python program to implement Decision Tree.

### **Aim**

To implement a Decision Tree classification algorithm using Python and train the model with given sample data. The program accepts new input values from the user and predicts the corresponding class using the trained decision tree.

### **Algorithm**

1. Start the program.
2. Create the input features and corresponding target classes.
3. Create a Decision Tree classifier using Scikit-learn.
4. Train the classifier using the given dataset.
5. Provide a new sample as input.
6. Predict the class of the sample using the trained model.
7. Display the input and predicted class.
8. Use the `plot_tree()` function to visualize the trained Decision Tree.
9. Display the feature names and class names in the graph.
10. □ Stop the program.

### **Program**

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
```

```
X = [[1,1],[1,0],[0,1],[0,0]]
y = [1,1,0,0]
```

```
m = DecisionTreeClassifier(max_depth=2,random_state=1)
m.fit(X,y)
```

```

x = [1,1]

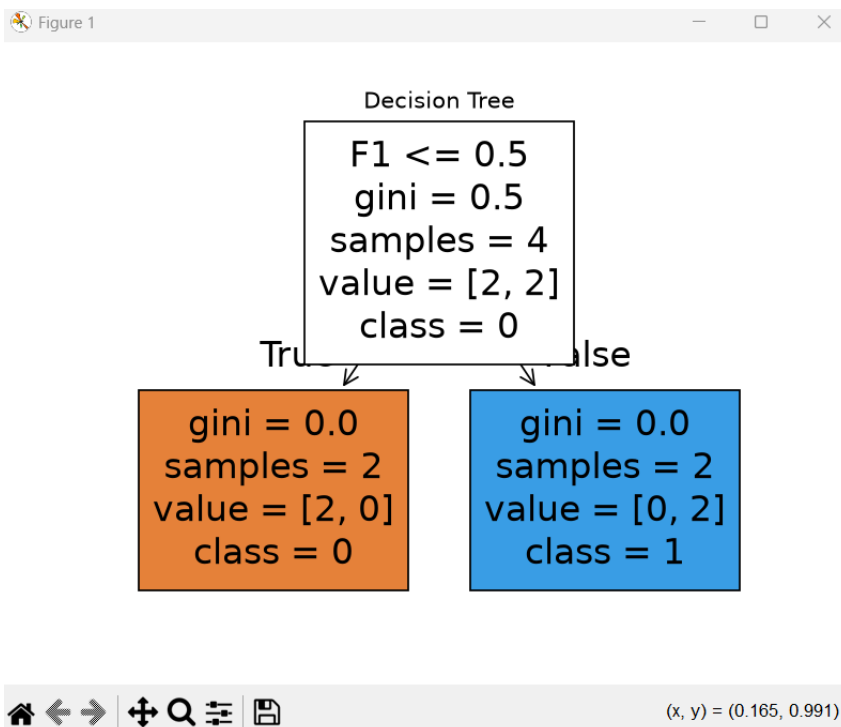
print("Input:",x)
print("Predicted Class:",m.predict([x])[0])

plot_tree(m,
          feature_names=["F1","F2"],
          class_names=["0","1"],
          filled=True)

plt.title("Decision Tree")
plt.show()

```

## Output



## Result:

The Decision Tree classification algorithm was successfully implemented using Python. The model was trained using the given feature values and class labels, and it successfully predicted the class of new user-provided input using the learned decision rules.